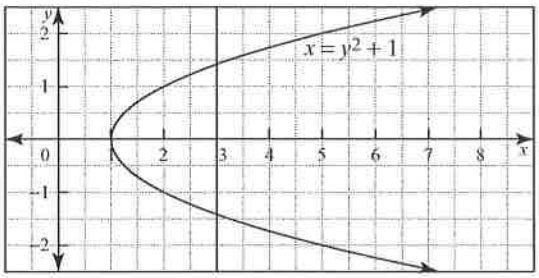
Chapter 1 – Functions – REVIEW
MCR3U
Jensen

SOLUTIONS

Section 1: Functions, Domain, and Range

1) State the domain and range of the following relations and then state if it is a function or not.

a)

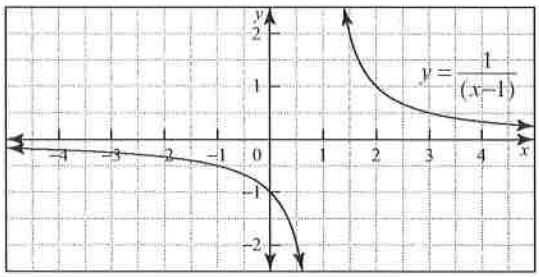


D: $\{x \in \mathbb{R} \mid x \geq 1\}$

R: $\{y \in \mathbb{R}\}$

NOT a function

b)

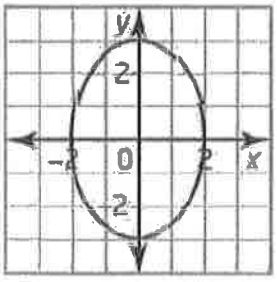


D: $\{x \in \mathbb{R} \mid x \neq 1\}$

R: $\{y \in \mathbb{R} \mid y \neq 0\}$

function

c)

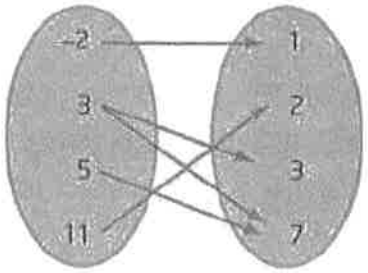


D: $\{x \in \mathbb{R} \mid -2 \leq x \leq 2\}$

R: $\{y \in \mathbb{R} \mid -3 \leq y \leq 3\}$

NOT a function

d)



D: $(-2, 3, 5, 11)$

R: $(1, 2, 3, 7)$

NOT a Function

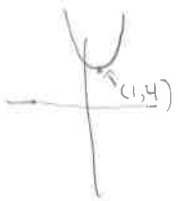
e) $\{(1, 4), (2, 6), (3, 10), (4, 18), (5, 29)\}$

D: $(1, 2, 3, 4, 5)$

R: $(4, 6, 10, 18, 29)$

Function

f) $y = 3(x - 1)^2 + 4$



D: $\{x \in \mathbb{R}\}$

R: $\{y \in \mathbb{R} \mid y \geq 4\}$

Function

Section 2: Max or Min of a Quadratic Function

2) Determine the vertex for each quadratic function by completing the square. State if the vertex is a maximum or a minimum.

a) $f(x) = x^2 - 10x + 7$

$= (x^2 - 10x) + 7$

$= (x^2 - 10x + 25 - 25) + 7$

$= (x^2 - 10x + 25) - 25 + 7$

$= (x - 5)^2 - 18$

Vertex is $(5, -18)$ → Min

b) $f(x) = x^2 + 2x + 6$

$= (x^2 + 2x + 1 - 1) + 6$

$= (x^2 + 2x + 1) - 1 + 6$

$= (x+1)^2 + 5$

Vertex is $(-1, 5)$ → min

c) $f(x) = 3x^2 - 30x + 73$

$= 3(x^2 - 10x) + 73$

$= 3(x^2 - 10x + 25 - 25) + 73$

$= 3(x^2 - 10x + 25) - 75 + 73$

$= 3(x - 5)^2 - 2$

Vertex is $(5, -2)$ → min

d) $f(x) = -2x^2 - 8x + 7$

$= -2(x^2 + 4x) + 7$

$= -2(x^2 + 4x + 4 - 4) + 7$

$= -2(x^2 + 4x + 4) + 8 + 7$

$= -2(x+2)^2 + 15$

Vertex is $(-2, 15)$ → Max

3) Use partial factoring to determine the vertex of each function. State if the vertex is a max or min.

a) $y = -x^2 + 4x + 11$

$11 = -x^2 + 4x + 11$

$0 = -x^2 + 4x$

$0 = -x(x - 4)$

$-1x = 0$ $\quad x - 4 = 0$

$x = 0$ $\quad x = 4$

$x\text{-vertex} = \frac{0+4}{2}$

$= 2$

$y\text{-vertex} = -(2)^2 + 4(2) + 11$

$= 15$

Vertex is $(2, 15)$ → Max

b) $y = 3x^2 - 18x + 14$

$14 = 3x^2 - 18x + 14$

$0 = 3x^2 - 18x$

$0 = 3x(x - 6)$

$3x = 0$ $\quad x - 6 = 0$

$x = 0$ $\quad x = 6$

$x\text{-vertex} = \frac{0+6}{2}$

$= 3$

$y\text{-vertex} = 3(3)^2 - 18(3) + 14$

$= -13$

Vertex is $(3, -13)$ → min

c) $y = 5x^2 + 14x - 21$

$0 = 5x^2 + 14x$

$0 = x(5x + 14)$

$x = 0$ $\quad 5x + 14 = 0$

$x = -14/5$

$x = -2.8$

$x\text{-vertex} = \frac{0+(-2.8)}{2}$

$= -1.4$

$y\text{-vertex} = 5(-1.4)^2 + 14(-1.4) - 21$

$= -30.8$

Vertex is $(-1.4, -30.8)$

Min

d) $y = -2x^2 - 11x + 1$

$0 = -2x^2 - 11x$

$0 = -1x(2x + 11)$

$-1x = 0$ $\quad 2x + 11 = 0$

$x = 0$ $\quad x = -11/2$

$x = -5.5$

$x\text{-vertex} = \frac{0+(-5.5)}{2}$

$= -2.75$

$y\text{-vertex} = -2(-2.75)^2 - 11(-2.75) + 1$

$= 16.125$

Vertex is $(-2.75, 16.125)$

Max

4) A hall charges \$30 per person for a sports banquet when 120 attend. For every 10 extra people that attend, the hall will decrease the price by \$1.50 per person. What number of people will maximize the revenue for the hall?

Revenue = (cost)(# of people)

$0 = (30 - 1.50x)(120 + 10x)$

$30 - 1.5x = 0 \qquad 120 + 10x = 0$

$x = 20 \qquad x = -12$

$x\text{-vertex} = \frac{20 + (-12)}{2}$

$= 4$

of people $= [120 + 10(4)]$

$= 160$ people.

5) The power, P, in watts, produced by a solar panel is given by the function $P(I) = -5I^2 + 100I$, where I represents the current, in amperes.

a) What value of the current will maximize the power? x-coord of vertex

$0 = -5I^2 + 100I$

$0 = -5I(I - 20)$

$-5I = 0 \qquad I - 20 = 0$

$I = 0 \qquad I = 20$

$x\text{-vertex} = \frac{0 + 20}{2}$

$= 10$

A current of 10 amps maximizes power.

) What is the maximum power?

$P(10) = -5(10)^2 + 100(10)$

$= 500$

The max power is 500 watts.

6) a) Find the vertex of the parabola defined by $f(x) = -\frac{1}{3}x^2 + 2x - 4$.

$f(x) = -\frac{1}{3}(x^2 - 6x) - 4$

$f(x) = -\frac{1}{3}(x^2 - 6x + 9 - 9) - 4$

$f(x) = -\frac{1}{3}(x^2 - 6x + 9) + 3 - 4$

$f(x) = -\frac{1}{3}(x - 3)^2 - 1$

Vertex is $(3, -1)$

b) Is the vertex a minimum or a maximum?

Max

c) Without finding them, how many x-intercepts does the parabola have? Explain.

None. Vertex is below x-axis and parabola opens down.

Section 3: Radicals

7) Perform each radical operation and simplify where needed.

a) $\sqrt{27} - 4\sqrt{3} + \sqrt{243} - 8\sqrt{81} + 2$

$= (\sqrt{9})(\sqrt{3}) - 4\sqrt{3} + (\sqrt{81})(\sqrt{3}) - 8(9) + 2$

$= 3\sqrt{3} - 4\sqrt{3} + 9\sqrt{3} - 72 + 2$

$= 8\sqrt{3} - 70$

b) $-3\sqrt{3}(\sqrt{3} + 5\sqrt{2})$

$= -3\sqrt{9} - 15\sqrt{6}$

$= -3(3) - 15\sqrt{6}$

$= -9 - 15\sqrt{6}$

c) $(5 + \sqrt{3})(5 - \sqrt{3})$ D.O.S.

$= (5)^2 - (\sqrt{3})^2$

$= 25 - 3$

$= 22$

d) $5\sqrt{2}(11 + 2\sqrt{2}) - 4(8 + 3\sqrt{2})$

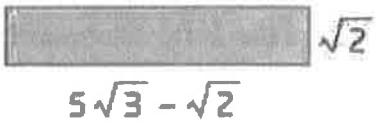
$= 55\sqrt{2} + 10\sqrt{4} - 32 - 12\sqrt{2}$

$= 43\sqrt{2} + 10(2) - 32$

$= 43\sqrt{2} - 12$

8) Find a simplified expression for the area of each shape

a)



$A = (\sqrt{2})(5\sqrt{3} - \sqrt{2})$

$= 5\sqrt{6} - 2$

b)



$A = \frac{\pi r^2}{2}$

$= \frac{\pi(3\sqrt{3})^2}{2}$

$= \frac{\pi(9\sqrt{9})}{2}$

$= \frac{27\pi}{2}$

Section 4: Solve Quadratics by Factoring

9) Solve each of the following quadratic equations by factoring.

a) $x^2 + 4x - 21 = 0$ S: 4 P: -21 (7 and -3)

$(x+7)(x-3) = 0$

$x+7=0 \quad x-3=0$

$x=-7 \quad x=3$

b) $5x^2 - 19x - 4 = 0$ S: -19 P: -20 (-20 and 1)

$5x^2 - 20x + 1x - 4 = 0$

$(5x^2 - 20x) + (1x - 4) = 0$

$5x(x-4) + 1(x-4) = 0$

$(x-4)(5x+1) = 0$

$x-4=0 \quad 5x+1=0$

$x=4 \quad x=-\frac{1}{5}$

c) $4x^2 + 12x = -9$ P.S.T.

$4x^2 + 12x + 9 = 0$

$(2x+3)^2 = 0$

$2x+3=0$

$x = -\frac{3}{2}$

Section 5: Solve Quadratics Using the Quadratic Formula

10) Solve each quadratic equation. Give exact answers.

a) $f(x) = 2x^2 - 9x - 1$

$x = \frac{9 \pm \sqrt{(-9)^2 - 4(2)(-1)}}{2(2)}$

$x = \frac{9 \pm \sqrt{89}}{4}$

$x = \frac{9+\sqrt{89}}{4}$ and $x = \frac{9-\sqrt{89}}{4}$

b) $g(x) = -3x^2 + 4x + 2$

$x = \frac{-4 \pm \sqrt{(4)^2 - 4(-3)(2)}}{2(-3)}$

$x = \frac{-4 \pm \sqrt{40}}{-6}$

$x = \frac{-4 \pm 2\sqrt{10}}{-6}$

$x = \frac{2(2 \mp \sqrt{10})}{-6}$

$x = \frac{2 - \sqrt{10}}{3}$ and $x = \frac{2+\sqrt{10}}{3}$

11) Solve each quadratic equation using any method.

a) $3x^2 - 15x = 42$

$3x^2 - 15x - 42 = 0$

$3(x^2 - 5x - 14) = 0$

$x^2 - 5x - 14 = 0$

$(x-7)(x+2) = 0$

$x - 7 = 0 \quad x + 2 = 0$

$x = 7 \quad x = -2$

b) $5x^2 + 11x + 1 = 0$

$x = \frac{-11 \pm \sqrt{(11)^2 - 4(5)(1)}}{2(5)}$

$x = \frac{-11 \pm \sqrt{101}}{10}$

$x = \frac{-11+\sqrt{101}}{10}$ and $x = \frac{-11-\sqrt{101}}{10}$

12) Use the discriminant to predict the number of real roots of...

a) $f(x) = x^2 + 3x + 3$

$b^2 - 4ac = (3)^2 - 4(1)(3)$

$= -3$

∴ No Roots

b) $f(x) = -2x^2 + 4x - 2$

$b^2 - 4ac = (4)^2 - 4(-2)(-2)$

$= 0$

∴ 1 Root.

c) $f(x) = 2x^2 + 5x - 8$

$b^2 - 4ac = (5)^2 - 4(2)(-8)$

$= 89$

∴ 2 Roots.

Section 6: Determine a Quadratic Equation Given its Roots

13) Determine the equation in standard form for each quadratic function...

a) x-intercepts $\overset{r}{-2}$ and $\overset{s}{5}$, containing the point $(\overset{x}{3}, \overset{y}{5})$

$y = a(x-r)(x-s)$

$5 = a(3+2)(3-5)$

$5 = a(5)(-2)$

$5 = -10a$

$a = -\frac{1}{2}$

$y = -\frac{1}{2}(x+2)(x-5)$

$y = -\frac{1}{2}(x^2-3x-10)$

$y = -\frac{1}{2}x^2 + \frac{3}{2}x + 5$

b) x-intercepts $-2 \pm \sqrt{5}$, containing the point (-4, 5)

$y = a(x-r)(x-s)$

$y = a[x-(-2+\sqrt{5})][x-(-2-\sqrt{5})]$

$y = a(x+2-\sqrt{5})(x+2+\sqrt{5})$ D.O.S.

$y = a[(x+2)^2 - (\sqrt{5})^2]$

$y = a(x^2+4x+4-5)$

$y = a(x^2+4x-1)$

sub in point (-4, 5)

$y = a(x^2+4x-1)$

$5 = a[(-4)^2+4(-4)-1]$

$5 = a(-1)$

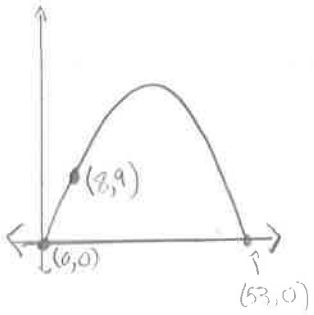
$a = -5$

$y = -5(x^2+4x-1)$

$y = -5x^2 - 20x + 5$

14) A golf ball is hit, and it lands at a point on the same horizontal plane 53 meters away. The path of the ball took it just over a 9 meter tall tree that was 8 meters in front of the golfer.

a) Assume the ball is hit from the origin of a coordinate plane. Find a quadratic function that describes the path of the ball.



$y = a(x-r)(x-s)$

$9 = a(8-0)(8-53)$

$9 = a(-360)$

$a = -\frac{1}{40}$

$y = -\frac{1}{40}(x)(x-53)$

$y = -\frac{1}{40}(x^2-53x)$

$y = -\frac{1}{40}x^2 + \frac{53}{40}x$

b) What is the maximum height of the ball?

$x\text{-vertex} = \frac{0+53}{2}$

$= 26.5$

$y\text{-vertex} = -\frac{1}{40}(26.5)^2 + \frac{53}{40}(26.5)$

≈ 17.6 m

Section 7: Linear-Quadratic Systems

15) Determine the points of intersection of each pair of functions.

a) $f(x) = 4x^2 - 15x + 20$ and $g(x) = 5x - 4$

$5x-4 = 4x^2-15x+20$

$0 = 4x^2-20x+24$

$0 = 4(x^2-5x+6)$

$0 = x^2-5x+6$

$0 = (x-2)(x-3)$

$x-2=0 \quad x-3=0$

$x=2 \quad x=3$

POI #1

$g(2) = 5(2)-4$

$= 6$

$(2,6)$

POI #2

$g(3) = 5(3)-4$

$= 11$

$(3,11)$

b) $f(x) = -2x^2 + 9x + 9$ and $g(x) = -3x - 5$

$-3x-5 = -2x^2+9x+9$

$0 = -2x^2+12x+14$

$0 = -2(x^2-6x-7)$

$0 = x^2-6x-7$

$0 = (x-7)(x+1)$

$x-7=0 \quad x+1=0$

$x=7 \quad x=-1$

POI #1

$g(7) = -3(7)-5$

$= -26$

$(7,-26)$

$g(-1) = -3(-1)-5$

$= -2$

$(-1,-2)$

16) For what value of b will the line $y = -2x + b$ be tangent to the parabola $y = 3x^2 + 4x - 1$

$-2x+b = 3x^2+4x-1$

$0 = 3x^2+6x-1-b$

$0 = 3x^2+6x+(-1-b)$

* set $b^2-4ac=0$ *

$(6)^2-4(3)(-1-b) = 0$

$36-4(-3-3b) = 0$

$36+12+12b = 0$

$48 = -12b$

$b = -4$

$\boxed{b=-4}$

Answers

1) a) domain: $\{X \in \mathbb{R} | x \geq 1\}$ range: $\{Y \in \mathbb{R}\}$ **b)** domain: $\{X \in \mathbb{R} | x \neq 1\}$ range: $\{Y \in \mathbb{R} | y \neq 0\}$

c) domain: $\{X \in \mathbb{R} | -2 \leq x \leq 2\}$ range: $\{Y \in \mathbb{R} | -3 \leq y \leq 3\}$

d) domain: $\{-2, 3, 5, 11\}$, range $\{1, 2, 3, 7\}$ **e)** domain: $\{1, 2, 3, 4, 5\}$, range: $\{4, 6, 10, 18, 29\}$

f) domain:$\{X \in \mathbb{R}\}$, range: $\{Y \in \mathbb{R} | y \geq 4\}$

2) a) (5, -18) min **b)** (-1, 5) min **c)** (5, -2) min **d)** (-2, 15) max

3) a) (2, 15) max **b)** (3, -13) min **c)** (-1.4, -30.8) min **d)** $\left(-\frac{11}{4}, \frac{129}{8}\right)$ max

4) 160

5) a) 10 A **b)** 500 W

6) a) (3, -1) **b)** max **c)** vertex is below x-axis and opens down, therefore no x-intercepts

7) a) $-70 + 8\sqrt{3}$ **b)** $-9 - 15\sqrt{6}$ **c)** 22 **d)** $-12 + 43\sqrt{2}$

8) a) $5\sqrt{6} - 2$ square units **b)** $\frac{27\pi}{2}$ square units

9) a) -7 and 3 **b)** $-\frac{1}{5}$ and 4 **c)** $-\frac{3}{2}$

10) a) $x = \frac{9+\sqrt{89}}{4}, x = \frac{9-\sqrt{89}}{4}$ **b)** $x = \frac{2-\sqrt{10}}{3}, x = \frac{2+\sqrt{10}}{3}$

11) a) $x = 7$ and $x = -2$ **b)** $x = \frac{-11 \pm \sqrt{101}}{10}$

12) a) none **b)** 1 **c)** 2

13) a) $f(x) = -\frac{1}{2}x^2 + \frac{3}{2}x + 5$ **b)** $f(x) = -5x^2 - 20x + 5$

14) a) $f(x) = -\frac{1}{40}x^2 + \frac{53}{40}x$ **b)** 17.6 m

15) a) (2, 6), (3, 11) **b)** (-1, -2), (7, -26)

16) $b = -4$